

THESE P AND I DIAGRAM IS DIAGRAMATIC ONLY. DESIGN OF PIPE LINES MUST BE INVESTIGATED FOR VENTING OF GAS AND VAPOR POCKETS IN PIPING AND EQUIPMENT. LOW POINTS IN PIPING, PUMPS AND EQUIPMENT FOR FREEZING AND DRAINING, AND ACCESSIBILITY OF ALL VALVES, FLANGES, INSTRUMENTS INCLUDING THERMOCOUPLES, ETC.

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14780-8120-25-23-0001G10000-23-25-23-0001G

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MOTOR CONTROL DETAILS

ITEM TAG NO.	P&ID NO.	MOTOR DETAILS	SIS SIGNAL CONNECTED	DCS SIGNAL CONNECTED	XI	XA (NOTE 6)	XR	XL	II	XP	HS (START)	HS (STOP)	HS(HOA)	TS
					RUNNING	ELECTRICAL FAULT	READY	PERMISSIVE	CURRENT	MOTOR POWER	IN DCS	IN DCS	IN FIELD	TEMP. SWITCH FOR MAGNETIC PUMP
P-2301A	14780-8120-25-23-0009	B			23-0901A	23-0901A		23-0901A	23-0901A		23-0901AST	23-0901ASP	23-0901AHA	
P-2301B	14780-8120-25-23-0009	B			23-0901B	23-0901B		23-0901B	23-0901B		23-0901BST	23-0901BSP	23-0901BHA	
P-2302A	14780-8120-25-23-0017	D		FY-23-1702	23-1701A	23-1701A	23-1701A	23-1701A	23-1701A		23-1701AST	23-1701ASP	23-1701AHA	
P-2302B	14780-8120-25-23-0017	D		FY-23-1702	23-1701B	23-1701B	23-1701B	23-1701B	23-1701B		23-1701BST	23-1701BSP	23-1701BHA	
P-2303A	14780-8120-25-23-0013	B			23-1301A	23-1301A		23-1301A	23-1301A		23-1301AST	23-1301ASP	23-1301AHA	
P-2303B	14780-8120-25-23-0013	B			23-1301B	23-1301B		23-1301B	23-1301B		23-1301BST	23-1301BSP	23-1301BHA	
P-2304A	14780-8120-25-23-0020A	D		LY-23-2001	23-2001A	23-2001A		23-2001A	23-2001A		23-2001AST	23-2001ASP	23-2001AHA	
P-2305A	14780-8120-25-23-0015	B			23-1501A	23-1501A		23-1501A	23-1501A		23-1501AST	23-1501ASP	23-1501AHA	
P-2305B	14780-8120-25-23-0015	B			23-1501B	23-1501B		23-1501B	23-1501B		23-1501BST	23-1501BSP	23-1501BHA	
P-2305C	14780-8120-25-23-0016	B			23-1601A	23-1601A		23-1601A	23-1601A		23-1601AST	23-1601ASP	23-1601AHA	
P-2305D	14780-8120-25-23-0016	B			23-1601B	23-1601B		23-1601B	23-1601B		23-1601BST	23-1601BSP	23-1601BHA	
P-2305E	14780-8120-25-23-0016	B			23-1602A	23-1602A		23-1602A	23-1602A		23-1602AST	23-1602ASP	23-1602AHA	
P-2305F	14780-8120-25-23-0016	B			23-1602B	23-1602B		23-1602B	23-1602B		23-1602BST	23-1602BSP	23-1602BHA	
P-2306A	14780-8120-25-23-0019	B			23-1901A	23-1901A		23-1901A	23-1901A		23-1901AST	23-1901ASP	23-1901AHA	
P-2306B	14780-8120-25-23-0019	B			23-1901B	23-1901B		23-1901B	23-1901B		23-1901BST	23-1901BSP	23-1901BHA	
P-2307A	14780-8120-25-23-0008B	D		FY-23-0802	23-0801A	23-0801A	23-0801A	23-0801A	23-0801A		23-0801AST	23-0801ASP	23-0801AHA	
P-2307B	14780-8120-25-23-0008B	D		FY-23-0802	23-0801B	23-0801B	23-0801B	23-0801B	23-0801B		23-0801BST	23-0801BSP	23-0801BHA	
P-2308A	14780-8120-25-23-0005A	L	UC-2301	FY-23-0502	23-0501A	23-0501A	23-0501A	23-0501A	23-0501A	23-0501A	23-0501AST	23-0501ASP	23-0501AHA	TSH-23-0508
P-2308B	14780-8120-25-23-0005A	L	UC-2301	FY-23-0502	23-0501B	23-0501B	23-0501B	23-0501B	23-0501B	23-0501B	23-0501BST	23-0501BSP	23-0501BHA	TSH-23-0509
P-2309A	14780-8120-25-23-0007A	L	UC-2301	FY-23-0703	23-0701A	23-0701A	23-0701A	23-0701A	23-0701A	23-0701A	23-0701AST	23-0701ASP	23-0701AHA	TSH-23-0704
P-2309B	14780-8120-25-23-0007A	L	UC-2301	FY-23-0703	23-0701B	23-0701B	23-0701B	23-0701B	23-0701B	23-0701B	23-0701BST	23-0701BSP	23-0701BHA	TSH-23-0705
P-2320	14780-8120-25-23-0020A	D		LS-23-2005	23-2001	23-2001		23-2001	23-2001		23-2001ST	23-2001SP	23-2001HA	
P-2316A	14780-8120-25-23-0010A	F	UC-2303		23-1001A	23-1001A		23-1001A	23-1001A		23-1001AST	23-1001ASP	23-1001AHA	
P-2316B	14780-8120-25-23-0010A	F	UC-2303		23-1001B	23-1001B		23-1001B	23-1001B		23-1001BST	23-1001BSP	23-1001BHA	
P-2317A	14780-8120-25-23-0010A	F	UC-2303		23-1002A	23-1002A		23-1002A	23-1002A		23-1002AST	23-1002ASP	23-1002AHA	
P-2317B	14780-8120-25-23-0010A	F	UC-2303		23-1002B	23-1002B		23-1002B	23-1002B		23-1002BST	23-1002BSP	23-1002BHA	

GENERAL NOTES

NOTES

- HIGH AND LOW MOTOR POWER ALARMS AND SHUTDOWNS ARE PROVIDED FOR MAGNETIC SEALLESS PUMPS.
- SHUTDOWN CIRCUITRY SHALL BE DESIGNED TO STOP THE OPERATING PUMP FOR AN ABNORMAL CONDITION BY ANY OF THE LISTED SHUTDOWN VARIABLES.
- ALARM CIRCUITRY SHALL BE DESIGNED TO SERIES CONNECT ALL LISTED ALARM CONTACTS, FOR EACH PUMP SERVICE, TO ACTIVATE THE RELATED ALARM IN THE CONTROL CENTER.
- HARD WIRE ELECTRICAL FAULT ALARM FOR HV MOTOR.
- XY CAN BE FY, LY, PY AND UY.
- ELECTRICAL FAULTS STATUS WILL BE INDICATED ON DCS AS FOLLOWINGS ;
 - HV MOTORS
 - INSTANTANEOUS OVERCURRENT
 - AC TIME OVERCURRENT
 - EARTH FAULT
 - PHASE BALANCE
 - THERMAL OVERLOAD
 - STALL FAULT
 - NOTCHING
 - TRIP CIRCUIT SUPERVISION
 - LOCK OUT
 - UNDERVOLTAGE
 - WINDING THERMAL DEVICE
 - DIFFERENTIAL RELAY (ONLY FOR C-2201)
 - LV MOTORS
 - INSTANTANEOUS OVERCURRENT
 - AC TIME OVERCURRENT
 - EARTH FAULT
 - PHASE BALANCE
 - THERMAL OVERLOAD
 - STALL FAULT
 - UNDERVOLTAGE



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BASED ON

PIPING AND INSTRUMENT DIAGRAM
TYPICAL PUMP MOTOR CONTROL SCHEMATIC (2/2)
CONCENTRATION, DECOMPOSITION, AND
NEUTRALIZATION SECTION
UOP PHENOL PROCESS UNIT

Z1	AS BUILT	HSO	SWP	JSC	SWL	4/25/16
S4	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	10/15/14
S3	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	10/8/14
S2	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	9/16/14
S1	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	8/20/14
R2	ISSUED FOR CONSTRUCTION WITH HOLDS	HSO	SWP	JSC	SWL	4/4/14
R1	ISSUED FOR CONSTRUCTION WITH HOLDS	HSO	SWP	JSC	SWL	2/28/14
A2	APPROVED FOR DESIGN	HSO	SWP	JSC	SWL	2/7/14
REV.	DESCRIPTION	BY	CHK.	APPR.	PROJ. APPR.	DATE

PROJ. NO.

DWG. NO.

REV.

PEN

120117

Z1

PPCL

14780-8120-25-23-0001G

SCALE